Reliability-Aware RGB-D Terrain Clustering for Robotic Guide Dogs under Sensor Degradation

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Abstract

Robotic guide dogs must recognize terrain while camera and depth streams degrade. We present a reliability-aware RGB-D terrain clustering method with per-sample sensor-health estimation and a calibrated online readout. Per-sample reliability scores are computed from exposure, contrast, sharpness, depth validity, and depth smoothness; softmax weights define a weighted concatenation that is clustered by reliability-weighted multi-view K-means. Clusters are mapped to terrain classes by majority vote on a small labelled calibration set, and online samples are assigned to the nearest cluster centroid, so clustering drives the terrain label used by a semantic-adaptive particle filter (SAPF). Silhouette scores are used only for evaluation. On a corrected NYU Depth V2 pilot subset (17 images per class), the method reaches 71.1% clustering accuracy and 38.3% NMI on clean data and 60.0% accuracy at the most severe injected degradation, the best among the compared methods. On a 351-sample ROS/Gazebo street scene, the cluster readout obtains 77.8% under blocked cross-validation, 90.1 ± 1.0% online, and 77.6 ± 1.6% under mixed sensor degradation. A fair comparison including a fixed large-covariance PF shows clear gains over the standard PF, while SAPF is comparable to the large-covariance baseline; a shuffled-label control is worse on the clean run. The study is limited to a pilot NYU subset, kinematic Gazebo locomotion, a simulated exteroceptive pose, and in-scene calibration.

Keywords: Reliability-aware multi-view clustering, RGB-D terrain perception, Sensor degradation, Cluster-to-terrain calibration, Semantic-adaptive particle filter, Robotic guide dog

1. Introduction

Robotic guide dogs are expected to accompany visually impaired users through sidewalks, tactile paving, obstacles, and staircases. Unlike vehicles operating on structured roads, a quadrupedal guide dog must interpret fine-grained terrain semantics while maintaining a reliable 6-DoF pose estimate in the presence of locomotion-induced vibration, foot slip, and sensor degradation. A single sensing modality is fragile in this setting: vision is affected by overexposure, motion blur, and illumination changes, whereas depth or LiDAR measurements can become sparse or corrupted by multipath reflections and dropouts.

A fixed fusion rule is not sufficient in this setting. When one modality is severely degraded, naive concatenation or a fixed weighting can propagate corrupted features into the downstream estimator. Equally important, the semantic label produced by the perception front end is rarely connected to the process noise of the state estimator: a conventional particle filter uses the same motion covariance on flat ground and on stairs, which can cause delayed elevation response or z-axis drift.

This paper therefore treats terrain clustering as the primary problem and state estimation as a downstream validation of the cluster output. The proposed front end, called reliability-aware adaptive heterogeneous physical feature fusion (RA-AHPFF), differs from a conventional silhouette-weighted pipeline in three respects. First, the modality weights are computed from per-sample sensor-health indicators rather than from silhouette scores, which removes the circular dependency between clustering and weighting. Second, the two descriptor blocks are combined by weighted

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concatenation in a common standardized space; PCA projections from different sensor spaces are not added. Third, the online terrain label is obtained by assigning the current descriptor to a cluster centroid and reading out the terrain class of that cluster, so that clustering is actually used by the robot rather than being an offline ablation only.

The main contributions are:

- We formulate a reliability-aware multi-view clustering front end with per-sample visual and depth health indicators, softmax modality weights, weighted concatenation, and reliability-weighted multi-view K-means. The weights are computed independently of the clustering objective, and silhouette scores are used only for evaluation.
- We connect clustering to the online system through a cluster-to-terrain calibration step. A small labelled set is used only to name the clusters; online samples are assigned to the nearest centroid and the mapped terrain label is published to the state estimator. A supervised classifier is retained only as an upper-bound comparison.
- We provide a fair downstream evaluation of the semantic-adaptive particle filter against odometry, a standard particle filter, a fixed large-covariance particle filter, an extended Kalman filter, an oracle-label variant, and a shuffled-label variant. The comparison is repeated over multiple runs and uses a reduced-rate exteroceptive pose stream with outage windows rather than an idealized 50 Hz absolute pose.
- We report the performance and the limitations of the resulting system on a corrected NYU Depth V2 pilot subset, a 351-sample in-scene RGB-D calibration set, and a ROS/Gazebo Go1 street scene. The code and the processed data are released to support reproduction.

2. Related Work

2.1. Sensor-Driven Assistive Navigation and Guide Dog Robots

Sensor-driven assistive technologies have been developed around smartphones, electronic white canes, and wearable devices [1, 2, 3]. The Virtual Guide Dog uses onboard sensors and the NTCIP protocol to communicate with traffic controllers at intersections [4]. With the progress of legged robotics, quadrupedal platforms have been explored as physical guide dogs. The iDog system uses AI and edge computing for real-time audio alerts [5], and Hwang et al. developed a guide dog robot with a rigid harness and a semantics-aware local planner [6]. Physical human-robot interaction and locomotion comfort are also critical: Fan et al. proposed a force-compliance MPC framework for interactive navigation [7], and Yu et al. emphasized quiet and stable locomotion for acoustic situational awareness [8]. These systems demonstrate the importance of robust perception and state estimation, but they typically use deterministic or loosely coupled fusion and do not connect terrain semantics to the process noise of a probabilistic filter.

2.2. Multi-View Clustering and Reliability-Aware Fusion

Multi-view clustering (MVC) provides a natural formulation for combining heterogeneous sensor views [9, 10, 11]. Graph-regularized nonnegative matrix factorization embeds local geometric structure and orthogonal constraints [12]; spectral analysis with local refinement captures fine-grained visual topology [13]; and angular metric learning can produce tighter class boundaries [14]. Continual and incomplete-view methods update consensus coefficients or handle missing samples [15, 16], and auto-weighted binary MVC learns view and sample weights jointly [17]. Most of these methods are designed for offline or soft-real-time clustering and assume that each view is available at test time. In a robotic system, the reliability of a view changes from frame to frame, so a fixed view weight is not sufficient. Our approach follows the reliability-weighting idea but keeps the descriptor set small and interpretable, and it explicitly calibrates clusters to terrain classes so that the result can be used online.

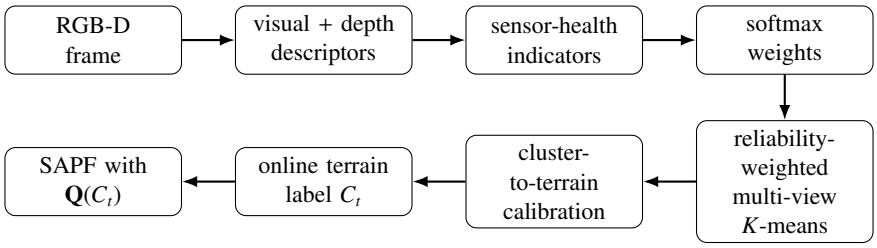


Figure 1: Overview of the revised pipeline. The health indicators define per-sample modality weights; clustering is performed in the weighted descriptor space; the cluster-to-terrain mapping supplies the online label used by the SAPF.

2.3. Semantic-Aware State Estimation

Particle filters are widely used for non-linear, multi-modal state estimation [18]. In legged and assistive robots, process and observation noise are often tuned for a single terrain type, which is inadequate when the robot transitions between flat ground, tactile paving, obstacles, and stairs. Semantic-aware estimation addresses this by using a terrain label to modulate the motion or observation model [**?** **?**]. Our SAPF follows this idea in a deliberately simple form: a discrete terrain label changes the diagonal process covariance, and the filter uses systematic resampling with an effective-sample-size trigger. We do not claim a new filtering algorithm; the purpose of the filter study is to test whether the cluster output is useful downstream and to compare it with strong fixed-covariance baselines.

3. Methodology

The framework consists of a reliability-aware clustering front end and a 6-DoF SAPF back end (Fig. 1). The front end produces a terrain label C_t; the back end uses C_t to select the process covariance. The front end is trained once from a calibration set and then runs online.

3.1. Interpretable RGB-D Descriptors

Let $\mathbf{X}_t^{(v)}$ and $\mathbf{X}_t^{(d)}$ denote the visual and depth observations. We extract interpretable descriptors rather than learned embeddings. The visual descriptor contains Canny edge density, Laplacian variance, mean and standard deviation of intensity, mean and standard deviation of gradient magnitude, intensity entropy, and, in the online RGB version, mean RGB values, mean saturation, and yellow/red fractions. The depth descriptor contains mean, standard deviation, minimum, maximum, 25th and 75th percentiles, depth span, mean absolute Sobel responses, mean gradient magnitude, and the ratio of valid depth pixels in the region of interest.

For NYU Depth V2, descriptors are computed inside the annotated object mask. For the RGB-D robot, they are computed inside a fixed lower-central region of interest $(0.15, 0.85) \times (0.35, 0.98)$ in normalized image coordinates. The visual block has 7 dimensions in the NYU experiment and 13 dimensions online; the depth block has 11 dimensions. The two blocks are standardized separately with statistics fitted on the calibration set.

3.2. Per-Sample Sensor Health and Modality Weights

The key change relative to silhouette-weighted fusion is that the weights depend on observable sensor quality, not on the clustering result. The visual health score is

$$s_v = 0.32e + 0.24c + 0.20h + 0.12g + 0.06\epsilon + 0.06\kappa, \tag{1}$$

where $e = 1 - 2|\mu_I - 0.5|$ is an exposure score, $c = \min(\sigma_I/0.28, 1)$ is contrast, $h = L/(L+40)$ with Laplacian variance L is sharpness, $g = \min(\bar{G}/0.20, 1)$ is texture strength, ϵ is normalized intensity entropy, and κ is normalized edge density. The visual reliability is $q_v = \sigma(5(s_v - 0.45))$, where $\sigma(\cdot)$ is the logistic function.

The depth health score is

$$s_d = 0.50r + 0.25\omega + 0.15\rho + 0.10\gamma, \tag{2}$$

where $r \in [0, 1]$ is the valid-pixel ratio, $\omega = 1/(1 + 6\bar{G}_d/\bar{D})$ is a smoothness score from the mean depth gradient $\bar{G}_d$ and mean depth $\bar{D}$, ρ is a normalized depth-spread score, and $\gamma = 1 - \exp(-\bar{D}/0.6)$ is a range score. The depth

reliability is $q_d = \sigma(5(s_d - 0.45))$. The coefficients are fixed from the sensor model and are not fitted to the terrain labels.

Given q_v and q_d, the modality weights are

$$w_v = \frac{\exp(\lambda q_v)}{\exp(\lambda q_v) + \exp(\lambda q_d)}, \qquad w_d = 1 - w_v, \tag{3}$$

with $\lambda = 4$ in all experiments. Because both reliabilities lie in $(0, 1)$, the weights are well defined, non-negative, and sum to one. Unlike the previous silhouette ratio $s_1/(s_1 + s_2)$, this construction cannot produce a negative or unbounded weight.

The standardized descriptor blocks are combined by weighted concatenation:

$$\mathbf{z}_t = \left[\sqrt{w_v}\, \tilde{\mathbf{z}}_t^{(v)},\ \sqrt{w_d}\, \tilde{\mathbf{z}}_t^{(d)} \right], \tag{4}$$

where $\tilde{\mathbf{z}}^{(v)}$ and $\tilde{\mathbf{z}}^{(d)}$ are the per-view standardized descriptors. The blocks remain in their own coordinate systems; no PCA score from one view is added to a PCA score from the other view.

3.3. Reliability-Weighted Multi-View K-Means

For a calibration set of n samples, let $\mathbf{z}_i^{(v)}$ and $\mathbf{z}_i^{(d)}$ be the standardized views, and let w_{vi}, w_{di} be their weights. We minimize the weighted within-cluster objective

$$J = \sum_{k=1}^{K} \sum_{i \in C_k} \left(w_{vi} \|\mathbf{z}_i^{(v)} - \mathbf{c}_k^{(v)}\|^2 + w_{di} \|\mathbf{z}_i^{(d)} - \mathbf{c}_k^{(d)}\|^2 \right), \tag{5}$$

where C_k is the set of samples assigned to cluster k. The assignment step uses the weighted distance in (5), and the centroid update is the weighted mean

$$\mathbf{c}_k^{(v)} = \frac{\sum_{i \in C_k} w_{vi} \mathbf{z}_i^{(v)}}{\sum_{i \in C_k} w_{vi}}, \qquad \mathbf{c}_k^{(d)} = \frac{\sum_{i \in C_k} w_{di} \mathbf{z}_i^{(d)}}{\sum_{i \in C_k} w_{di}}. \tag{6}$$

This is equivalent to running K-means on the weighted concatenation (4) while keeping the centroid update consistent with the per-sample weights.

The number of clusters is written as $K = rC$, where C is the number of terrain classes and r is an over-clustering factor. In the unsupervised NYU benchmark we use $r = 1$ so that the clustering result is directly comparable to the number of annotated classes. In the calibrated online readout we use $r = 8$ for the four-class RGB-D deployment ($K = 32$) and $r = 3$ for the three-class pilot readout ($K = 9$). Over-clustering captures within-class appearance modes, such as different floor textures or stair viewpoints; each cluster is still mapped to a single terrain class by the calibration step below. The values of r are calibration hyperparameters and are not selected using the evaluation labels.

3.4. Cluster-to-Terrain Calibration and Online Readout

Let ℓ_i be the cluster index of calibration sample i and y_i its terrain class. For each cluster k, the class is chosen by majority vote:

$$m_k = \arg \max_{c \in \{1,\dots,C\}} \sum_{i:\ell_i = k} \mathbf{1}[y_i = c]. \tag{7}$$

Clusters without labelled samples are assigned to the global majority class. This step uses labels only to name clusters; it does not train a discriminative classifier.

At run time, the current descriptor is standardized with the stored per-view scalers, its reliability weights are computed from (3), and the sample is assigned to the nearest centroid under the weighted distance. The published terrain label is $\hat{C}_t = m_{\hat{k}}$, where $\hat{k}$ is the assigned cluster. The confidence is the softmax of the negative cluster distances. A short majority filter over the last five predictions suppresses isolated label flips. A supervised ExtraTrees classifier is retained only as an upper-bound comparison in Section 4.2; it is not used in the online pipeline.

Table 1: Process-noise standard deviations used in (9). Positions are in metres and angles in radians. The diagonal variance is the sum of the squared base and class-specific standard deviations.

Quantity	x	y	z	roll	pitch	yaw
Base $\boldsymbol{\sigma}_{base}$	0.002	0.002	0.001	0.001	0.001	0.001
Flat $\boldsymbol{\sigma}_0$	0	0	0	0	0	0
Stairs $\boldsymbol{\sigma}_1$	0.002236	0.002236	0.005916	0.003873	0.010954	0.001732
Obstacle $\boldsymbol{\sigma}_2$	0.011832	0.011832	0.001732	0.005916	0.001732	0.010954
Tactile $\boldsymbol{\sigma}_3$	0.005	0.005	0.002	0.002	0.002	0.005

3.5. Semantic-Adaptive Particle Filter

The state is $\mathbf{x}_t = [p_x, p_y, p_z, \phi, \theta, \psi]^T$, where ϕ, θ, ψ are roll, pitch, and yaw. The filter maintains N particles $\{\mathbf{x}_t^{[i]}, w_t^{[i]}\}_{i=1}^N$.

3.5.1. Motion Model and Semantic Covariance

The state transition is

$$\mathbf{x}_t^{[i]} = \mathbf{x}_{t-1}^{[i]} + \mathbf{u}_t + \boldsymbol{\epsilon}_t(C_t), \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}(C_t)), \tag{8}$$

where $\mathbf{u}_t$ is the odometry increment. The process covariance is

$$\mathbf{Q}(C_t) = \mathrm{diag}\Big(\boldsymbol{\sigma}_{base}^2 + \boldsymbol{\sigma}_{C_t}^2\Big), \tag{9}$$

where $\boldsymbol{\sigma}_{base}$ is the nominal process-noise standard deviation and $\boldsymbol{\sigma}_{C_t}$ is a class-specific boost vector. Stairs increase the vertical and pitch components; obstacles increase the horizontal and yaw components; tactile paving receives a small increase. The numerical values are listed in Table 1. A standard particle filter (SPF) uses $\mathbf{Q}_{base} = \mathrm{diag}(\boldsymbol{\sigma}_{base}^2)$ for every label. A large-covariance particle filter (LPF) replaces $\boldsymbol{\sigma}_{C_t}$ with the elementwise maximum over all classes for every label; it is the strongest fixed-covariance baseline in our study.

3.5.2. Observation Model and Resampling

The observation is a 6-DoF pose measurement

$$\mathbf{z}_t = \mathbf{x}_t + \mathbf{n}_t, \quad \mathbf{n}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_t), \tag{10}$$

which is used to update the importance weights:

$$w_t^{[i]} = \eta\, p(\mathbf{z}_t \mid \mathbf{x}_t^{[i]})\, w_{t-1}^{[i]}. \tag{11}$$

The likelihood is Gaussian in position and orientation, with angles wrapped to $[-\pi, \pi]$. To avoid an idealized observation model, the exteroceptive pose is available at 5 Hz, not at the 50 Hz filter rate, and two 4 s outage windows (8–12 s and 18–22 s) remove the pose update. Between updates, the filter relies on the motion model; this is the regime in which the process covariance matters. In the standalone simulation the outage windows are the same, and the observation standard deviation is 0.05 m in z on flat ground and 0.10 m on stairs. The remaining base observation standard deviations are 0.02 m in x and y, 0.01 rad in roll and yaw, and 0.02 rad in pitch on stairs; the simulator uses the same values in the Gazebo study.

The effective sample size is

$$\mathrm{ESS}_t = \frac{1}{\sum_{i=1}^N (w_t^{[i]})^2}. \tag{12}$$

When $\mathrm{ESS}_t < N/2$, systematic resampling is applied. With $u_0 \sim \mathcal{U}[0, 1/N)$ and $u_i = u_0 + i/N$, particle i is selected according to the cumulative weight interval containing u_i. The state estimate uses the weighted mean for positions and the circular mean for angles.

Table 2: Clustering on the corrected NYU Depth V2 pilot subset (17 images per class, 20 seeds). ACC and NMI are percentages; mean±standard deviation over seeds. The severe setting is degradation level 2.0.

	Clean		**Severe degradation**		**Severe, other metrics**	
Method	ACC	NMI	ACC	NMI	ARI	Silhouette
Vision only	46.2±3.1	8.0±2.0	42.5±4.0	5.5±4.0	−0.005	0.432
Depth only	65.0±0.7	33.2±1.1	59.6±3.5	**27.1±4.5**	0.194	0.345
Fixed concat	56.6±2.0	24.7±4.1	54.4±5.2	19.1±5.0	0.121	0.246
PCA average	68.5±3.4	35.8±3.5	49.2±6.0	11.2±6.8	0.060	0.104
Uniform weighted	56.7±1.7	24.8±3.6	54.2±5.3	18.7±5.5	0.121	0.248
Shuffled reliability	66.7±3.6	33.9±3.3	56.2±3.3	21.2±4.2	0.148	0.215
RA-AHPFF (Ours)	**71.1±2.1**	**38.3±2.4**	**60.0±4.7**	23.2±5.1	**0.195**	0.208

3.6. ROS/Gazebo Integration

The Go1 is equipped with a forward RGB-D camera (Fig. 7). The implementation uses ROS [**?**] and Gazebo [**?**]. The perception node extracts the RA-AHPFF descriptors from the color and depth images, vertically flips the raw camera image in software, computes the health weights, assigns the sample to a cluster centroid, and publishes the mapped terrain label at approximately 10 Hz. The SAPF node runs at 50 Hz with 500 particles, reads the robot pose from Gazebo model states, simulates noisy leg odometry and a 5 Hz exteroceptive pose with outages, and publishes the filtered pose. The street world contains a concrete sidewalk, curb and road, brick facade, tactile paving, a ten-step staircase (0.15 m rise, 0.30 m run), a bollard, street furniture, a static pedestrian, a walking pedestrian actor, and a scripted human handler. The Go1 model state is commanded along a scripted trajectory; its legs are animated, but the motion is not generated by a physics-based locomotion controller. The human handler is a gravity-disabled, non-colliding model that is teleported with the robot. These choices bound the interpretation of the Gazebo results.

4. Experiments

4.1. NYU Depth V2 Pilot Clustering

We use the verified NYU label ids floor=11, stairs=215, and chair=5 (obstacle proxy). The ids 1/42/4, which are sometimes used by mistake, correspond to book/shelves/ceiling in the 1-based NYU annotation and are not terrain classes. NYU Depth V2 [**?**] contains only 17 frames with at least 100 annotated stair pixels; the balanced subset therefore contains 17 images per class (51 images in total). This is a pilot subset, not a large benchmark.

We extract object-masked descriptors and evaluate clean data and three degradation levels. At each level, half of the samples (stratified by class) are visually degraded and the other half are depth-degraded, so each degraded sample loses exactly one modality. Visual degradation combines overexposure, reduced contrast and sharpness, and feature noise; depth degradation combines pixel dropout, reduced depth span, increased gradient noise, and feature noise. The level controls the severity. We repeat each experiment with 20 random seeds and report mean±standard deviation. The number of clusters is $K = 3$, the number of annotated classes. All methods use the same majority-vote cluster-to-class mapping. Silhouette scores are reported only as a diagnostic.

Table 2, Table 3, and Fig. 2 report the results. RA-AHPFF is best on clean data in both ACC and NMI (71.1% and 38.3%), and has the best ACC at the most severe degradation (60.0%, with 23.2% NMI). At mild degradation, depth only remains strong (63.5% and 61.0% at levels 1.0 and 1.5), which is expected because the NYU depth channel is highly informative for stairs and floors. The advantage of RA-AHPFF is largest when the dataset contains both visually degraded and depth-degraded samples and when the fixed PCA average degrades sharply. The bootstrap 95% confidence interval of the accuracy at level 1.5 is [45.0, 70.6] for RA-AHPFF and [43.1, 70.6] for depth only, so the difference at that level should not be over-interpreted on a 51-image pilot.

The reliability weights behave as intended. On clean data the mean weights are $w_v = 0.281$ and $w_d = 0.719$. On visually degraded samples the mean visual weight falls to 0.205, 0.108, and 0.100 at levels 1.0, 1.5, and 2.0,

Table 3: Clustering accuracy (%) versus degradation level on the NYU pilot subset. Each level uses 20 seeds; the values are means.

Method	Clean	Level 1.0	Level 1.5	Level 2.0
Vision only	46.2	44.9	44.7	42.5
Depth only	65.0	63.5	61.0	59.6
Fixed concat	56.6	59.3	55.9	54.4
PCA average	68.5	55.2	52.5	49.2
Uniform weighted	56.7	59.6	55.4	54.2
Shuffled reliability	66.7	60.1	59.1	56.2
RA-AHPFF (Ours)	**71.1**	60.4	**60.2**	**60.0**

Table 4: In-scene terrain readout on the 351-sample Go1 calibration set. The cluster readout uses $K = 32$ clusters ($r = 8$) and a majority-vote cluster-to-terrain mapping. The supervised ExtraTrees model is an upper-bound comparison. Per-class recall is from the blocked protocol.

Readout	Random CV	Blocked CV	Flat	Stairs	Obstacle	Tactile
Cluster readout (RA-AHPFF)	91.2	77.8	0.44	0.87	0.79	0.87
Supervised ExtraTrees	97.7	80.9	0.69	0.88	0.76	0.82

respectively. On depth-degraded samples the mean depth weight falls from 0.719 to 0.503, 0.345, and 0.298. The shuffled-reliability control uses the same reliability values but permutes their sample correspondence; it is consistently worse than the proposed weighting at levels 1.5 and 2.0, which shows that the correspondence between weights and sample quality matters, not just the presence of larger weights.

A silhouette sweep on the 51 images selected $K = 2$, whereas the benchmark uses $K = 3$. This disagreement is a symptom of the small pilot: the floor and obstacle classes form a broad continuum in this subset. We therefore treat $K = 3$ as a class-count prior rather than as evidence of a stable unsupervised model order, and we do not claim that the pilot supports fully unsupervised model selection.

4.2. In-Scene Cluster Readout and Online Accuracy

We evaluate the calibrated readout on 351 RGB-D samples collected in the Go1 street scene: Flat 59, Stairs 152, Obstacle 85, and Tactile 55. The samples are calibrated on the same scene, so this experiment measures the in-scene closed loop, not cross-domain generalization. We use two cross-validation protocols. Random 5-fold cross-validation mixes adjacent frames and therefore leaks appearance information across folds. Blocked 5-fold cross-validation sorts each class by timestamp and assigns contiguous time blocks to folds; this is the more conservative estimate.

Table 4 shows that random cross-validation is optimistic for both readouts (91.2% and 97.7%), whereas the blocked protocol gives 77.8% for the cluster readout and 80.9% for the supervised model. The cluster readout retains most of the supervised blocked accuracy without training a discriminative classifier; the remaining gap is concentrated in the Flat class, which is visually similar to some obstacle and tactile samples. This is the main reason we report blocked cross-validation in the paper and treat the random result only as a leakage diagnostic.

In the online closed loop, the cluster readout reaches $90.1 \pm 1.0\%$ time-aligned semantic accuracy over three 30 s runs. This number is higher than the blocked cross-validation estimate because the deployed model is fitted on all 351 samples from the same scene, including frames from the same temporal neighbourhood; it should not be read as a cross-scene result. Under overexposure, horizontal motion blur, and 50% depth dropout from 6 to 20 s, the online accuracy falls to $77.6 \pm 1.6\%$. Fig. 8 shows a degraded preview.

4.3. Reliability Weights under Modality-Specific Degradation

To verify that the weights respond to the degraded modality rather than to the terrain class alone, we ran four sessions with the same calibration model and the same trajectory: baseline, overexposure, motion blur, and depth dropout. The degradation is active from 6 to 20 s. Table 5 reports the mean weights and qualities during that window.

Fig. 3 and Table 5 show the same pattern. The visual weight drops from 0.425 at baseline to 0.174 under overexposure and to 0.338 under motion blur, while the depth weight rises correspondingly. Under depth dropout the pattern

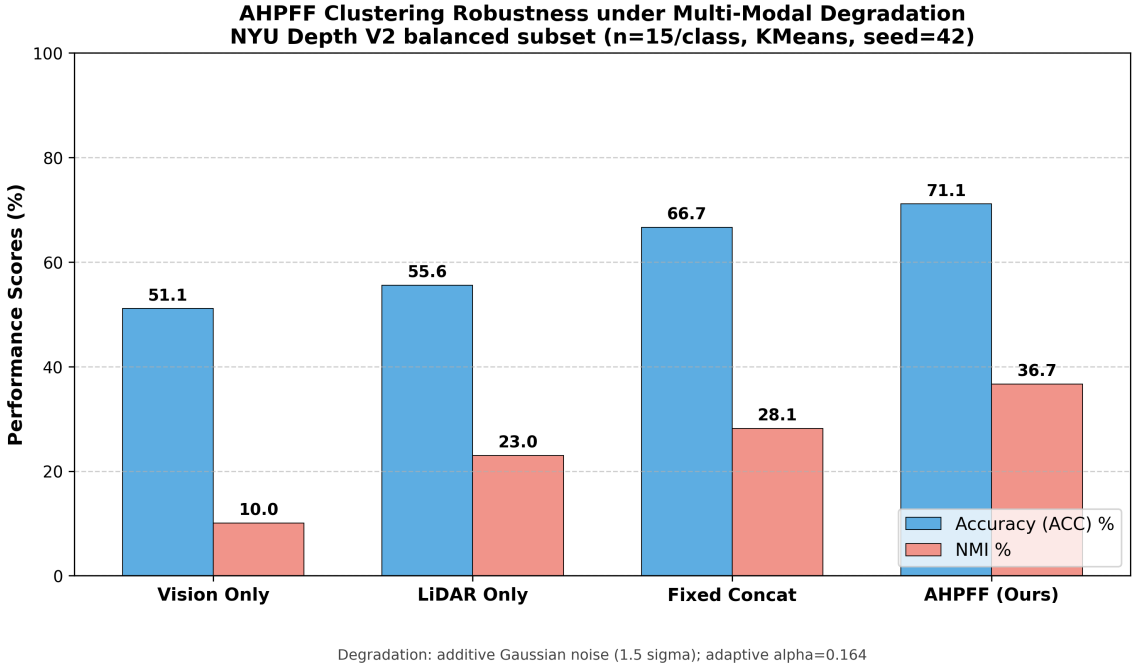


Figure 2: Clustering on the corrected NYU Depth V2 pilot subset. Left: accuracy at the severe degradation level (2.0). Right: accuracy versus degradation level. The proposed reliability weighting is best on clean data and at the most severe level; depth-only clustering remains strong at mild degradation.

Table 5: Mean modality weights and health qualities during the 6–20 s degradation window in the Go1 street scene. The weights are computed per sample from (3), not set globally.

Condition	w_v	w_d	q_v	q_d
Baseline	0.425	0.575	0.760	0.839
Overexposure	0.174	0.826	0.442	0.844
Motion blur	0.338	0.662	0.668	0.843
Depth dropout	0.724	0.276	0.761	0.513

reverses: the depth weight falls to 0.276 and the visual weight rises to 0.724. The health qualities explain the weights: overexposure reduces q_v from 0.760 to 0.442, and depth dropout reduces q_d from 0.839 to 0.513. This is direct evidence that the per-sample reliability design responds to the degraded modality; a fixed weight or a silhouette-derived weight cannot produce this behavior.

4.4. Semantic-Adaptive Particle Filter: Fair Comparison

We first evaluate the filters on a standalone 30 s stair-climbing simulation with 20 seeds. The trajectory contains flat ground, an obstacle, and a staircase; odometry has a small vertical bias on stairs. The semantic labels come from the three-class cluster readout fitted on the corrected NYU pilot set (holdout accuracy 66.7%, sequence semantic accuracy 73.0%). The exteroceptive pose is available at 5 Hz with the same two outage windows as the Gazebo study. The compared methods are leg odometry (LO), the standard particle filter (SPF), the fixed large-covariance particle filter (LPF), an extended Kalman filter (EKF), SAPF with predicted labels, SAPF with oracle labels, and SAPF with shuffled predicted labels. All particle filters use 1000 particles and the same odometry and observation streams.

Fig. 4 and Table 6 show the standalone results, which are deliberately sobering. The large fixed covariance LPF is a very strong baseline (0.1107 m position RMSE), and the predicted-label SAPF is close to it (0.1158 m). The oracle-label SAPF is slightly better (0.1123 m), and the shuffled-label SAPF is clearly worse (0.1374 m). The semantic label

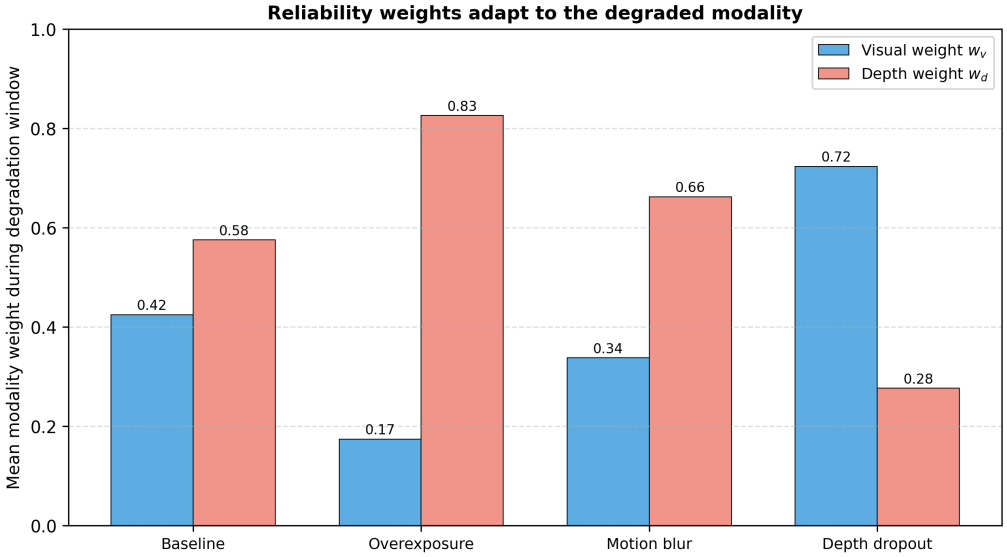


Figure 3: Mean modality weights during the degradation window in the Go1 street scene. The visual weight decreases under overexposure and blur, while the depth weight decreases under depth dropout.

Table 6: Standalone 6-DoF stair-climbing simulation over 20 seeds (1000 particles). Values are mean RMSE in metres (orientation in radians). The sequence semantic accuracy of the cluster readout is 73.0%.

Method	***Z* RMSE**	**Position RMSE**	**Orientation RMSE**	**Stairs *Z* RMSE**
LO	0.2960	0.3456	0.0935	0.3342
SPF	0.2597	0.2647	0.0291	0.3141
LPF	**0.0992**	**0.1107**	0.0283	0.1478
EKF	0.1670	0.1734	0.0257	0.2476
SAPF-Pred	0.1062	0.1158	0.0264	0.1552
SAPF-Oracle	0.1014	0.1123	0.0268	0.1426
SAPF-Shuffled	0.1295	0.1374	0.0261	0.1920

therefore carries information, but a globally inflated covariance can recover much of the benefit when the exteroceptive pose is reasonably accurate. We do not claim that semantic adaptation alone explains the improvement over the standard PF.

Fig. 5 and Fig. 6 show a representative trajectory and label stream. We then repeat the closed-loop Go1 experiment three times with different random seeds, using the same 5 Hz observation schedule and outages. Table 7 reports the mean and standard deviation. The semantic accuracy is 90.1 ± 1.0% at baseline and 77.6 ± 1.6% under the mixed degradation. The SPF remains poor in both conditions (0.1711 m and 0.1726 m overall), while the LPF and SAPF are both around 0.09 m. The SAPF is comparable to the LPF at baseline (0.0877 m versus 0.0883 m) and slightly worse under degradation (0.0958 m versus 0.0937 m). The shuffled-label control is worse at baseline (0.1131 m) and close to the SAPF under degradation (0.0999 m), which indicates that the terrain label is useful mainly when the label stream is reliable.

The oracle-label SAPF is not uniformly the best in Table 7. This is not a contradiction: the filter performance depends on when the uncertainty is inflated, not only on whether the class is correct at the current instant. A label stream that anticipates a transition can be more useful than an exact but late label, and the predicted stream is smoothed over five frames. We therefore use the shuffled-label control, not the oracle, as the main evidence that terrain timing matters; the control preserves the label distribution but destroys the temporal correspondence.

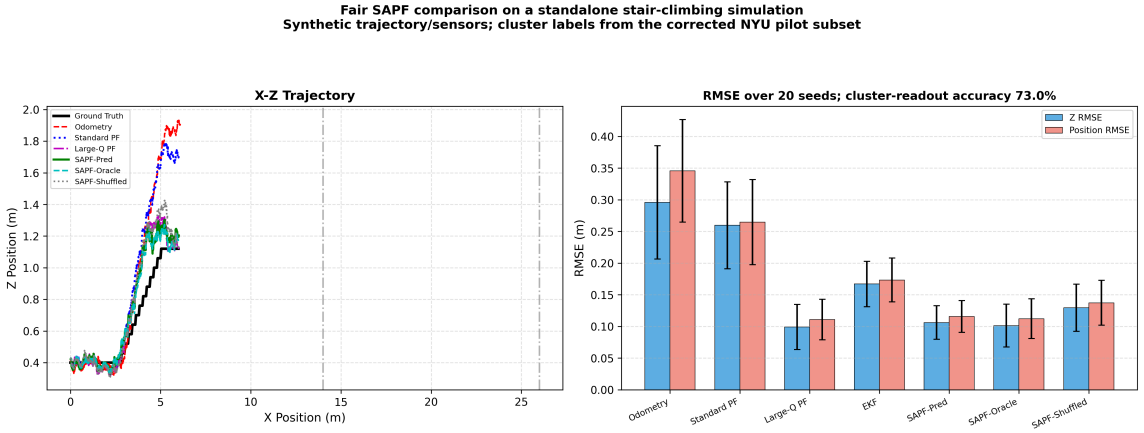


Figure 4: Standalone SAPF comparison over 20 seeds. The large-covariance PF is a strong baseline; the predicted-label SAPF is close to it, and the shuffled-label control is worse.

Table 7: Go1 closed-loop pose RMSE over three 30 s runs. Values are mean±standard deviation in metres. The exteroceptive pose is available at 5 Hz with two 4 s outages.

	Baseline		**Degraded**	
Method	Overall	Z	Overall	Z
LO	0.2520±0.0607	0.1861±0.0684	0.2592±0.0649	0.1819±0.0824
SPF	0.1711±0.0313	0.1593±0.0350	0.1726±0.0289	0.1607±0.0337
LPF	0.0883±0.0078	**0.0640±0.0049**	**0.0937±0.0107**	**0.0721±0.0120**
EKF	0.1285±0.0113	0.1153±0.0122	0.1267±0.0110	0.1131±0.0152
SAPF-Pred	**0.0877±0.0080**	0.0663±0.0088	0.0958±0.0127	0.0762±0.0151
SAPF-Oracle	0.1015±0.0098	0.0829±0.0071	0.1137±0.0157	0.0962±0.0197
SAPF-Shuffled	0.1131±0.0031	0.0979±0.0080	0.0999±0.0134	0.0830±0.0158

4.5. Discussion and Limitations

The experiments support a narrow claim: reliability-aware clustering is a practical way to obtain online terrain labels for a guide-dog robot when the camera or depth stream degrades, and the resulting labels are useful to a downstream filter. The evidence also shows the limits of the claim. First, the NYU study is a 51-image pilot subset; the confidence intervals are wide and depth-only clustering is competitive at mild degradation. Second, the in-scene readout is calibrated on the same Gazebo scene, and the blocked cross-validation estimate (77.8%) is lower than the online accuracy (90.1%) precisely because the online model sees the whole scene. Third, the large-covariance PF is a strong baseline; semantic adaptation is comparable rather than dramatically better, and the oracle-label variant is not uniformly best. Fourth, the Gazebo observation is a simulated exteroceptive pose at 5 Hz with outages, and the Go1 motion is kinematic; the study does not demonstrate a physics-based gait or a real-world deployment. Fifth, the health coefficients and the over-clustering factors are fixed calibration parameters, not learned end-to-end. Sixth, the online readout still confuses the Flat class with visually similar obstacle and tactile samples, as the blocked per-class recall in Table 4 shows.

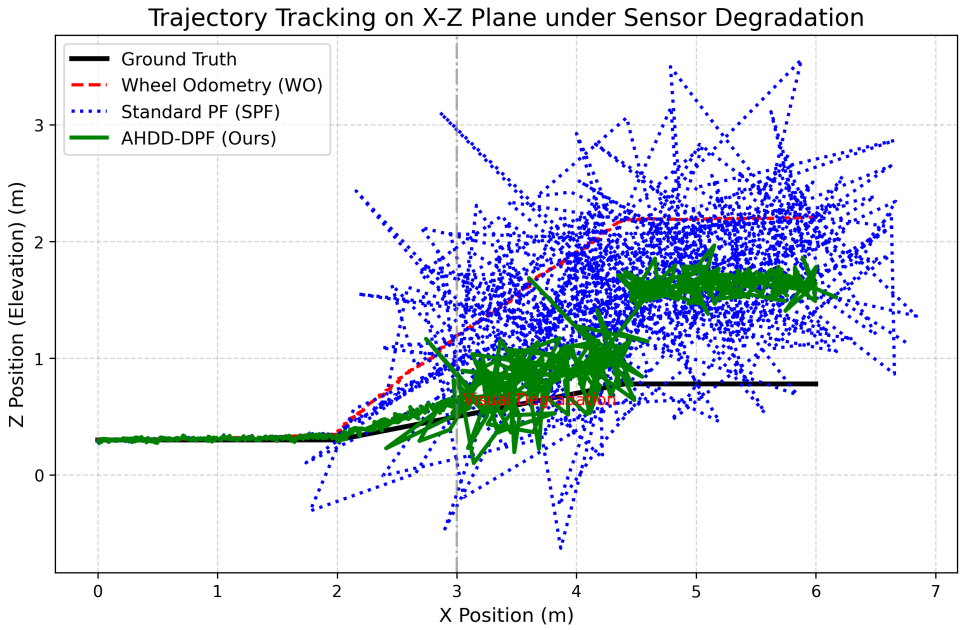


Figure 5: X–Z trajectory in the Go1 closed-loop study. The standard PF lags on the staircase; the large-covariance PF and SAPF both track the elevation, and the shuffled-label SAPF drifts more than the predicted-label SAPF.

5. Conclusion

We presented a reliability-aware RGB-D terrain clustering front end for robotic guide dogs and connected it to a 6-DoF semantic-adaptive particle filter. The method replaces silhouette-derived fusion weights with per-sample sensor-health weights, uses weighted concatenation instead of adding PCA coordinates from different sensor spaces, and calibrates cluster indices to terrain classes so that clustering is used online. On a corrected NYU pilot subset, the method has the best clean accuracy and the best accuracy at severe degradation, and the weights respond correctly to visually and depth-degraded samples. In a Go1 Gazebo street scene, the cluster readout reaches 90.1% online accuracy and 77.8% under blocked cross-validation, compared with 80.9% for a supervised upper bound; under mixed sensor degradation the online accuracy falls to 77.6%, while the SAPF remains close to a strong large-covariance baseline and clearly improves over the standard particle filter. The study is limited by the small NYU subset, kinematic Gazebo locomotion, a simulated exteroceptive pose, and in-scene calibration. Future work will (i) evaluate the method on a larger multi-scene RGB-D terrain dataset, including outdoor LiDAR data such as SemanticKITTI [**?**], (ii) integrate a physics-based locomotion controller, and (iii) test the cluster readout on a physical quadrupedal platform.

Acknowledgments

This work was supported by the Special Environment Robotics Laboratory at Southwest University of Science and Technology.

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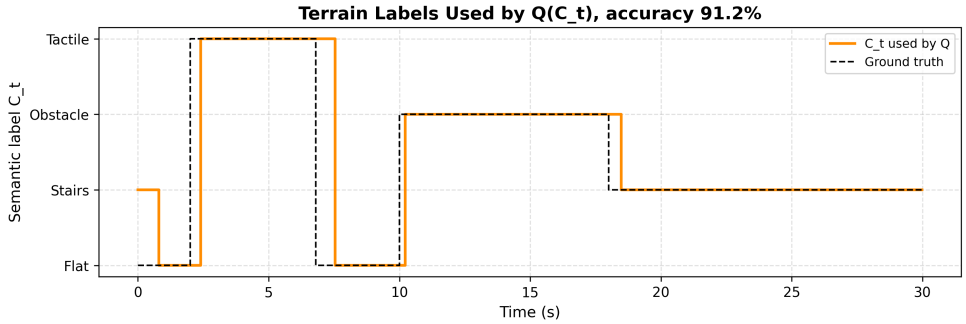


Figure 6: Terrain label C_t used by the SAPF versus the scripted ground-truth label in the Go1 baseline run. The online cluster readout reaches 90.1±1.0% time-aligned accuracy over three runs.

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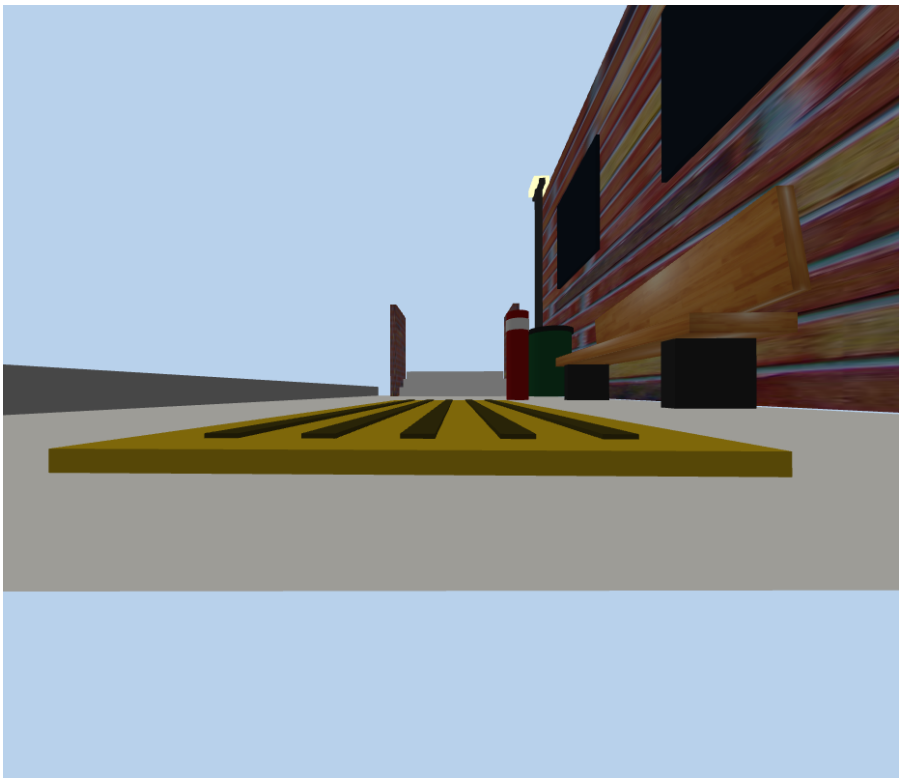


Figure 7: Rectified forward RGB-D view of the Go1 street scene. The scene contains tactile paving, a bollard, a walking pedestrian actor, street furniture, and a staircase.

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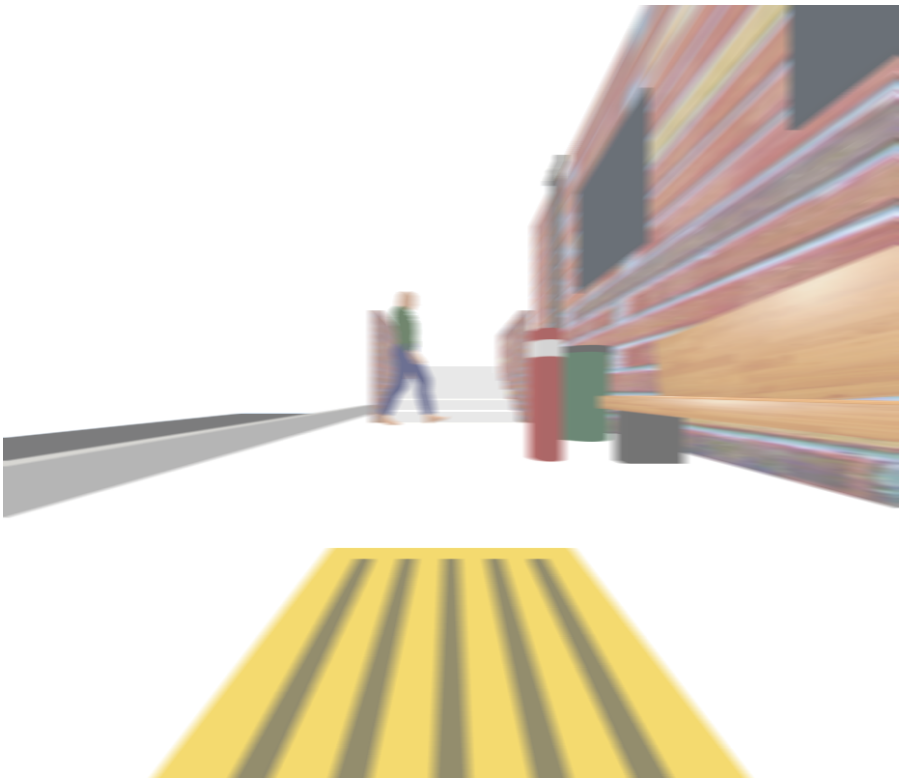


Figure 8: Degraded RGB-D preview used in the sensor-degradation study. The image shows overexposure and horizontal motion blur; the depth stream also loses 50% of its pixels during the degradation window.